

Mariia Zhytnikova *Full Stack / Software Engineer*

Location: Helsinki, Uusimaa, Finland

Phone: (+358) 46 57 60 200

Email: allusio@gmail.com

LinkedIn: [linkedin.com/in/mariia-zhytnikova](https://www.linkedin.com/in/mariia-zhytnikova)

GitHub: <https://github.com/MariiaZhytnikova/>

Languages: English – Fluent | Ukrainian – Native | Finnish – B1

SUMMARY

Programming student at Hive Helsinki with a background in science and hands-on experience from multiple projects. I've built a strong foundation in systems programming and enjoy working across different programming languages to solve technical problems.

I enjoy breaking down complex problems and troubleshooting challenging technical issues in both individual and team-based projects. My research background strengthened my communication skills, ability to work under pressure, and to meet tight deadlines. I'm highly motivated to learn new technologies and continuously improve my skills.

Hive Helsinki's peer-to-peer learning environment strengthened my collaboration, adaptability, and problem-solving skills through intensive team projects, code reviews, and community-driven learning. I enjoy working in collaborative environments, sharing knowledge, and contributing to supportive developer communities.

TECHNICAL SKILLS

Programming Languages: TypeScript, C, C++, Python, Go

Backend Frameworks & APIs: Node.js, NestJS, Fastify, REST APIs, Socket.io, TypeORM, Swagger / OpenAPI, JWT

Frontend: React, Vite, REST API & WebSocket integration, Tailwind CSS, Styled-components

Databases: PostgreSQL, MariaDB, SQLite, SQL basics

Tools & DevOps: Git, GitHub, Docker, Docker Compose, Nginx, CI/CD (GitHub Actions), Jest

TEAM PROJECTS

Maze is Lava – Full-stack online turn-based board game

- **Frontend:** [React](#), [TypeScript](#), [Vite](#), [Socket.IO](#), and [REST APIs](#).

- Implemented lobby, join/create game flows, routing, settings, and authentication UI (guest & Google login)
- Built realtime UI updates for chat, friends presence, and game status
- Improved responsive UI and realtime state management patterns

- **Backend:** [NestJS](#), [PostgreSQL](#), [Socket.IO](#), [Docker](#), and [REST API](#) architecture.

- Developed core game engine systems including level registry, DTOs/models, and game creation/start flows
- Built realtime gateways for chat, friend presence, and live game updates
- Implemented persistence for games, leaderboards, and achievements

- Added backend tests and admin panel support

Hive — Browser-based turn-based strategy game **TypeScript, Canvas**

- Implemented game mechanics, rendering logic, player interactions, and responsive UI
- Developed minimax AI with alpha-beta pruning and heuristic evaluation
- Deployed browser-based multiplayer-ready game architecture

AirGuardian — **Python, JavaScript**

- Implemented telemetry processing and violation-detection logic for drone monitoring
- Built realtime visualization and services for telemetry data
- Integrated external JSON APIs and database persistence

Webserver — **C++**

- Built an HTTP/1.1 server inspired by Nginx using socket programming
- Implemented request parsing, routing, error handling, and cookie-based sessions
- Worked with low-level networking, concurrency concepts, and system programming

EDUCATION

Helsinki Hive, Helsinki — Software Development

Nov 2024 - 2026

Karazin University, Kharkiv — *Ph.D. in Biophysics, Mathematical & Physical Sciences, M.Sc. in Biophysics with honors*

June 2017

EXPERIENCE

Research officer

Institute of Radiophysics and Electronics NASU - Kharkiv, Ukraine

Dec 2017 - June 2024

- Worked with computational simulations, complex datasets, and scientific analysis workflows
- Automated data processing and interpreted large-scale research results

Senior Scientific Consultant

NIBR (external collaboration via IFNUL) - Lviv (remote), Ukraine

March 2018 - Apr 2024

- Conducted technical research and analytical reporting for biomedical projects
- Synthesized complex information into structured reports for decision-making

COURSES

Supervised Machine Learning: Regression and Classification (Stanford University / Coursera)

Learned to build and train supervised ML models using Python, NumPy, and scikit-learn.

OpenCV (IT Jim): Image processing, feature detection, object tracking, real-time video analysis.

Kaggle: Data preprocessing, model selection, performance evaluation through competitions.

Data Scientist (Data Quest): Python, pandas, data cleaning, visualization, statistical modeling.

Full Stack Open (Helsinki University) *ongoing*: JavaScript, TypeScript, Node.js, React, PostgreSQL, REST APIs.